

● HARD

● ALGEBRA

● ANSWER KEY

Linear equations in two variables

04

2 answers · Rationale-rich · Teach from doc

Q1**.5**

Accept also: 1/2

The correct answer is $\frac{1}{2}$. For an equation in slope-intercept form $y = mx + b$, m represents the slope of the line in the xy -plane defined by this equation. It's given that line p is defined by $4y + 8x = 6$. Subtracting $8x$ from both sides of this equation yields $4y = -8x + 6$. Dividing both sides of this equation by 4 yields $y = -\frac{8}{4}x + \frac{6}{4}$, or $y = -2x + \frac{3}{2}$. Thus, the slope of line p is -2 . If line r is perpendicular to line p , then the slope of line r is the negative reciprocal of the slope of line p . The negative reciprocal of -2 is $-\frac{1}{-2} = \frac{1}{2}$. Note that $1/2$ and $.5$ are examples of ways to enter a correct answer.

Q2**74**

The correct answer is 74. The y -intercept of a line in the xy -plane is the ordered pair (x,y) of the point of intersection of the line with the y -axis. Since line k intersects the y -axis at the point $(0,-6)$, it follows that $(0,-6)$ is the y -intercept of this line. An equation of any line in the xy -plane can be written in the form $y = mx + b$, where m is the slope of the line and b is the y -coordinate of the y -intercept. Therefore, the equation of line k can be written as $y = mx + (-6)$, or $y = mx - 6$. The value of m can be found by substituting the x - and y -coordinates from a point on the line, such as $(2,2)$, for x and y , respectively. This results in $2 = 2m - 6$. Solving this equation for m gives $m = 4$. Therefore, an equation of line k is $y = 4x - 6$. The value of w can be found by substituting the x -coordinate, 20, for x in the equation of line k and solving this equation for y . This gives $y = 4(20) - 6$, or $y = 74$. Since w is the y -coordinate of this point, $w = 74$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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